LAHORE UNIVERSITY OF MANAGEMENT SCIENCES
Department of Electrical Engineering

AI 501 Mathematics for Artificial Intelligence
Quiz 6 Solutions

Name: _____

Campus ID: _____

Total Marks: 10
Time Duration: 20 minutes

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1. [4 marks] To ensure reliable IoT-enabled vending machines in Pakistan, manufacturers monitor machine defects through AI-based anomaly detection systems. Suppose three companies A, B, and C produce vending machines, and their market shares and defect rates are as follows:

   - **Company A:** Produces 80% of all vending machines and has a defect rate of 4%.
   - **Company B:** Produces 15% of all vending machines and has a defect rate of 6%.
   - **Company C:** Produces 5% of all vending machines and has a defect rate of 9%.

   An **AI model** has flagged a randomly selected vending machine as defective. We aim to determine the probability that this defective machine came from Company A

   Solution:

   $$\frac{(0.80)(0.04)}{(0.80)(0.04)+(0.15)(0.06)+(0.05)(0.09)} = 0.7033$$

2. [6 marks] In digital audio and image processing, a continuous-valued signal is often *quantized* to a finite set of levels. A common model for quantization error is *uniform noise*.

   Let a single clean sample (e.g., a pixel or audio sample) be denoted by $x_0 \in \mathbb{R}$. After uniform quantization with step size $\Delta > 0$, we model the observed (noisy) value as

   $$Y = x_0 + U,$$

   where U represents the quantization error and follows a uniform distribution given by

   $$U \sim \text{Unif}\Big[-\frac{\Delta}{2}, \frac{\Delta}{2}\Big]$$

   For fixed x_0 and Δ, view Y as a random variable over the noise U. Find the mean and variance of Y. What is the pdf of Y?

   Solution:

   We have

   $$Y = x_0 + U, \qquad U \sim \text{Unif}\Big[-\frac{\Delta}{2}, \frac{\Delta}{2}\Big].$$

   Recall that if $U \sim \text{Unif}[a, b]$, then

   $$\mathbb{E}[U] = \frac{a+b}{2}, \qquad \text{Var}(U) = \frac{(b-a)^2}{12}.$$

   Here $a = -\frac{\Delta}{2}$, $b = \frac{\Delta}{2}$, so

   $$\mathbb{E}[U] = 0, \qquad \text{Var}(U) = \frac{\Delta^2}{12}.$$

   For fixed x_0,

   $$\mathbb{E}[Y] = \mathbb{E}[x_0 + U] = x_0 + \mathbb{E}[U] = x_0,$$

and

$$\mathrm{Var}(Y) = \mathrm{Var}(x_0 + U) = \mathrm{Var}(U) = \frac{\Delta^2}{12}.$$

Since Y is a shifted uniform random variable,

$$Y \sim \mathrm{Unif}\Big[x_0 - \frac{\Delta}{2},\, x_0 + \frac{\Delta}{2}\Big].$$

Thus the pdf of Y is

$$p_Y(y) = \begin{cases} \dfrac{1}{\Delta}, & \text{if } y \in \big[x_0 - \frac{\Delta}{2},\, x_0 + \frac{\Delta}{2}\big], \\ 0, & \text{otherwise.} \end{cases}$$